



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE
(AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		II / IV - B.Tech. I - Semester							
ELECTRICAL & ELECTRONICS ENGINEERING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23BS2102	Numerical Methods and Complex Variables	BS	3	0	0	3	30	70	100
B23HS2101	Universal Human Values -II : Understanding Harmony and Ethical Human Conduct	HS	2	1	0	3	30	70	100
B23EE2101	Electrical Circuit Analysis – II	PC	2	1	0	3	30	70	100
B23EE2102	DC Machines & Transformers	PC	2	1	0	3	30	70	100
B23EE2103	Analog Electronic Circuits	PC	3	0	0	3	30	70	100
B23EE2104	Electrical Machines - I Laboratory	PC	0	0	3	1.5	30	70	100
B23EE2105	Analog Electronic Circuits Laboratory	PC	0	0	3	1.5	30	70	100
B23EE2106	Circuit Simulation Laboratory	SEC	0	1	2	2	30	70	100
B23MC2102	Environmental Science	MC	2	0	0	-	30	-	30
TOTAL			14	4	8	20	270	560	830

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23BS2102	BS	3	--	--	3	30	70	3 Hrs.
NUMERICAL METHODS AND COMPLEX VARIABLES								
(For EEE)								
Course Objectives: Students are expected to								
1.	Acquire Computational Skills through numerical methods so that they can handle various real-world applications.							
2.	Familiarize the powerful tools of analyticity, residue calculus of complex functions and conformal mappings.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Find an approximate root of algebraic or transcendental equations using a few numerical algorithms and choose an appropriate interpolation formula to interpolate or extrapolate the given data with equal and unequal intervals.							K3
2.	Apply certain numerical techniques in evaluating definite integrals and find the approximate solution of certain Initial-Value- problems (IVP's) that arouse in Engineering applications.							K3
3.	Apply the concepts of analyticity and General Cauchy's integral formula to evaluate complex integrals.							K3
4.	Determine the Taylor and Laurent series of complex functions by using the nature of Isolated singularities and indenting the contours that convert complex integrals into simpler integrals by calculus of residues.							K3
5.	Solve the problems in Engineering applications by a powerful tool of conformal mapping either by changing the domain or the behavior of a function which is essential in designing and analysis.							K3
SYLLABUS								
UNIT-I (10Hrs)	UNIT – I: Iterative Methods. Introduction, Solution of algebraic and transcendental equations: Bisection method, Method of false position, Iteration method, Secant method and Newton-Raphson method. Interpolation: Newton's forward and backward interpolation formulae for interpolating the data with equi-spaced intervals and Lagrange's interpolation formula for data with unequal intervals.							
UNIT-II (10 Hrs)	UNIT – II: Numerical integration and Solution of ordinary differential equations with initial conditions. Trapezoidal rule, Simpson's one-third rule and three-eighth rule. Solution of initial value problems by Taylor's method, Picard's method, Euler's method,							

	Runge-Kutta method (second and fourth order), Milne's Predictor and Corrector Method.
UNIT-III (10 Hrs)	UNIT – III: Functions of a complex variable and Complex integration. Introduction, Continuity, Differentiability, Analyticity, Cauchy-Riemann equations in Cartesian and polar coordinates, Harmonic and conjugate harmonic functions, Milne – Thomson method. Complex integration: Line integral, Cauchy's integral theorem, Cauchy's integral formula – Generalized integral formula (all without proofs) and problems.
UNIT-IV (10 Hrs)	UNIT – IV: Series expansions and Residue Theorem. Radius of convergence, Expansion of function in Taylor's series, Maclaurin's series and Laurent series. Singularity, Isolated Singularity, Types of Isolated Singularities: Removable, Pole of order m and Essential, Residues, Residue theorem (without proof) – Evaluation of real definite integrals of the type $\int_c^{c+2\pi} f(\cos \theta, \sin \theta) d\theta$ and $\int_{-\infty}^{\infty} f(x) dx$.
UNIT-V (10 Hrs)	UNIT – V: Conformal mapping. Geometrical representation of the complex function, Some standard transformations: Translation, Rotation, Inversion and Bilinear transformation, Properties of Bi-Linear Transformation, fixed point and critical point. Conformal transformation, Special conformal mappings $w = z^2, z^n$ (n , a positive integer), $(z + a/z), e^z, \ln z, \sin z, \cos z$.
Textbooks:	
1.	B. S. Grewal , Higher Engineering Mathematics, 44 th Edition, Khanna Publishers.
2.	Micheael Greenberg , Advanced Engineering Mathematics, 2 nd edition, Pearson edition.
Reference Books:	
1.	Erwin Kreyszig , Advanced Engineering Mathematics, 10 th Edition, Wiley-India.
2.	B. V. Ramana , Higher Engineering Mathematics, 2007 Edition, Tata Mc. Graw Hill Education.
3.	Steven C. Chapra , Applied Numerical Methods with MATLAB for Engineering and Science, Tata Mc. Graw Hill Education.
4.	M. K. Jain, S.R.K. Iyengar and R.K. Jain , Numerical Methods for Scientific and Engineering Computation, New Age International Publications.
5.	J. W. Brown and R. V. Churchill , Complex Variables and Applications, 9 th edition, Mc-Graw Hill, 2013.
e-Resources:	
1.	https://archive.nptel.ac.in/courses/127/106/127106019/
2.	https://archive.nptel.ac.in/courses/111/106/111106141/

Course Code	Category	L	T	P	C	CIE	SEE	Exam
B23HS2101	HS	2	1	--	3	30	70	3 Hrs.
UNIVERSAL HUMAN VALUES-II: UNDERSTANDING HARMONY AND ETHICAL HUMAN CONDUCT								
(Common to all Programmes of Engineering)								
Course Objectives: The objective of this course is to make the student aware of								
1	Essential complementarity between 'Values' and 'Skills' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.							
2	Harmony in the human being, family, society and nature/existence							
3	Holistic perspective towards life, profession and happiness							
Course Outcomes: At the end of this course student will be able to								
S. No.	Outcome							Knowledge Level
1	Explain the role of value education in achieving basic human aspirations.							K2
2	Summarize needs to obtain harmony in self(I).							K2
3	Describe criteria for human-human relationship and harmony in society							K2
4	Explain four orders of nature and our existence							K2
5	Interpret significance of harmony in holistic development							K2
SYLLABUS								
UNIT-I (9 Hrs)	Introduction to Value Education: Understanding Value Education- Need, Basic Guidelines, Content and Process for Value Education Purpose and motivation for the course. Self-exploration as the Process for Value Education - Sharing about Oneself. Myers-Briggs Type Indicator (MBTI) Personality Test. Continuous Happiness and Prosperity – the Basic Human Aspirations and their Fulfilment. Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) - Exploring Human Consciousness. Happiness and Prosperity – Current Scenario. Method to Fulfil the Basic Human Aspirations - Exploring Natural Acceptance-understanding and living in harmony at various levels.							
UNIT-II (9 Hrs)	Harmony in the Human Being: Understanding Human being as the Co-existence of the Self and the Body. Distinguishing between the Needs of the Self and the Body - Exploring the difference of Needs of Self (I) and Body (Happiness and Physical Facility). The Body as an Instrument of the Self (I)’ (I being the doer, seer and enjoyer). Understanding Harmony in the Self(I) - Exploring Sources of Imagination in the Self(I). Harmony of the Self (I) with the Body (characteristics and activities of ‘I’ and harmony in ‘I’).							

	Programme to ensure self-regulation(<i>Sanyam</i>) and Health(<i>Swasth</i>)- Exploring Harmony of Self (I) with the Body.
UNIT-III (9 Hrs)	Harmony in the Family and Society: Harmony in the Family – the Basic Unit of Human Interaction. 'Trust' – the Foundational Value in Relationship- Exploring the Feeling of Trust- intention and competence. 'Respect' – as the Right Evaluation- Exploring the Feeling of Respect. Other Feelings, Justice in Human-to-Human Relationship. Understanding Harmony in the Society- (society being an extension of family). Vision for the Universal Human Order- Exploring Systems to fulfil Human Goal.
UNIT-IV (6 Hrs)	Harmony in the Nature/Existence: Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four orders of nature- recyclability and self-regulation in nature. Realizing Existence as Co-existence at All Levels - Understanding Existence as Co-existence of mutually interacting units in all pervasive space. Holistic perception of harmony at all levels of existence.
UNIT-V (9 Hrs)	Implications of the Holistic Understanding – a Look at Professional Ethics: Natural acceptance of human values. Definitiveness of (ethical) human conduct. A Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics: a. Ability to utilize the professional competence for augmenting universal human order b. Ability to identify the scope and characteristics of people friendly and eco-friendly production systems, c. Ability to identify and develop appropriate technologies and management patterns for above production systems. Holistic technologies, production systems and management models- typical case studies. Strategies for transition towards value based life and profession (from the present state to Universal Human Order): a. At the level of individual: as socially and ecologically responsible engineers, technologists and managers, b. At the level of society: as mutually enriching institutions and organizations.
Text Books	
1.	R R Gaur, R Sangal, G P Bagaria. "Human Values and Professional Ethics", Excel Books, New Delhi, 2010
2.	R R Gaur, R Asthana, G P Bagaria. "Teachers' Manual for A Foundation Course in Human Values and Professional Ethics", 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-53-2
Reference Books:	
1.	Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.
2.	Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3.	The Story of Stuff (Book).
4.	The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi
5.	Small is Beautiful - E. F Schumacher
6.	Slow is Beautiful - Cecile Andrews

7.	Economy of Permanence - J C Kumarappa
8.	Bharat Mein Angreji Raj – Pandit Sunderlal
9.	Rediscovering India - by Dharampal
10.	Hind Swaraj or Indian Home Rule - by Mohandas K. Gandhi
11.	India Wins Freedom - Maulana Abdul Kalam Azad
12.	Vivekananda - Romain Rolland (English)
e-Resources	
1.	https://fdp-si.aicte-india.org/UHV%20II%20Teaching%20Material/UHV%20II%20Lecture%2023-25%20Ethics%20v1.pdf
2.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Note.php
3.	https://fdp-si.aicte-india.org/download/FDPTeachingMaterial/3-days%20FDP-SI%20UHV%20Teaching%20Material/Day%203%20Handouts/UHV%203D%20D3-S2A%20Und%20Nature-Existence.pdf



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE2101	PC	2	1	--	3	30	70	3 Hrs.
ELECTRICAL CIRCUIT ANALYSIS-II								
(For EEE)								
Course Objectives: Student will learn								
1.	The transient behavior of RL, RC and RLC circuits.							
2.	The application of Laplace transforms to electrical circuits.							
3.	The concept of network functions, poles, zeros and filters.							
4.	The calculation of Two port network parameters.							
Course Outcomes: At the end of the course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Analyze the transient behavior of RL and RC circuits by solving describing differential equations.							K4
2.	Analyze the transient behavior of RLC circuits by solving describing differential equations.							K4
3.	Determine the complete response in electrical circuits using Laplace transform technique.							K3
4.	Illustrate the concept of network functions, poles, zeros and filters.							K3
5.	Find the two port network parameters of electrical networks.							K3
SYLLABUS								
UNIT-I (10Hrs)	DC Transient Analysis-First order circuits: Inductance and Capacitance transient behavior, Introduction to first order circuits, Source-free RC circuit, Source-free RL circuit, Evaluation of initial conditions, Step Response of an RC circuit, Step Response of an RL circuit, Concepts of Natural, Forced and Complete Response.							
UNIT-II (10 Hrs)	DC Transient Analysis-Second order circuits: Introduction to Second order circuits, Source- free series RLC circuit, Source- free parallel RLC circuit, Step Response of a series RLC circuit, Step Response of a parallel RLC circuit, The lossless LC circuit.							
UNIT-III (10 Hrs)	Laplace Transform Techniques for Transient Analysis: The Laplace transformation, Definition and Basic theorems of Laplace transformation, Initial and Final value theorems, Inverse Laplace transforms, Laplace transform of typical signals, Laplace transformation application to simple circuits considering unit step and							

	sinusoidal forcing functions.
UNIT-IV (10 Hrs)	Network Functions and Filters: Network Functions: The concept of Complex frequency, Transform impedance and transform circuits, Network functions for the one port and two ports, poles and Zeroes of Network Functions. Filters: Introduction to filters, Frequency response, Elementary treatment of passive low pass, high pass, band pass and band stop filters.
UNIT-V (10 Hrs)	Two port network parameters: Relationship of Two-Port Variables, Short circuit Admittance (Y)-Parameters, Open circuit Impedance (Z)-Parameters, Hybrid (H)-Parameters, Transmission (ABCD)-Parameters, two port parameters for resistive and simple AC Networks, Relationships between Parameter sets, Series, Parallel and Cascade connection of two port networks.
Textbooks:	
1.	Engineering Circuit Analysis, William Hayat Jr. and Jack E. Kemmerley, 8 th Edition, McGraw Hill International Edition.
2.	Network Analysis, 3 rd edition, M.E. Van Valkenburg, Pearson Education, Inc
Reference Books:	
1.	Fundamentals of Electric circuits, K. Alexander and Matthew Sadiku, 6 th Edition, McGraw Hill International Edition.
2.	Circuit Theory Analysis and Synthesis., 7 th Edition 2014 by Abhijit Chakrabarti, Dhanpat Rai & Co.
e-Resources	
1.	https://nptel.ac.in/courses/117106108
2.	https://nptel.ac.in/courses/108105159

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE2102	PC	2	1	--	3	30	70	3 Hrs.
DC MACHINES & TRANSFORMERS								
(For EEE)								
Course Objectives: Students will learn about								
1.	The principle and construction of DC generator, characteristics of DC generators, Armature reaction and commutation.							
2.	The working principle, operation & speed control of DC Motors and testing of DC Machines.							
3.	The construction, operating principle of single-phase transformer, phasor diagrams, losses and efficiency.							
4.	The operation of Auto transformer and testing of single-phase transformer.							
5.	The poly - phase connections, vector groups and Parallel operation of three-phase transformers.							
Course Outcomes: At the end of the course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Apply the concepts of electromechanical energy conversion to understand the principle and performance of DC generator.							K3
2.	Analyze the performance of DC machines using various tests.							K4
3.	Calculate the parameters of equivalent circuit, efficiency and regulation of single-phase transformer.							K3
4.	Analyze the performance of Single-phase transformer using various tests and understand the construction & Operation of Auto transformer.							K4
5.	Explain the poly - phase connections, vector groups, Scott connection and Parallel operation of three-phase transformers.							K3
SYLLABUS								
UNIT-I (10Hrs)	DC Generators: Introduction to electromechanical energy conversion- Construction and principle operation of DC machines – EMF equation for generator –Methods of Excitation – characteristics of DC generators –applications of DC Generators– Armature reaction and commutation							
UNIT-II (10 Hrs)	DC Motor and Testing of DC Machines: Principle operation of DC Motor, Back-emf and Torque equations of DC motor Types of DC motors, Characteristics of DC motors – losses and efficiency – applications of DC motors. Necessity of a starter – starting by 3-point starter – Speed control by armature voltage and field current control methods– Testing of DC machines – brake test, Swinburne’s test –Hopkinson’s test.							

UNIT-III (10 Hrs)	Single-phase Transformers: Introduction to single-phase Transformers (Construction and principle of operation)–emf equation – operation on no-load and on load –lagging, leading and unity power factors loads –phasor diagrams– equivalent circuit –regulation – losses and efficiency — All day efficiency.
UNIT-IV (10 Hrs)	Autotransformer and Testing of Transformers: Auto transformer construction and operation- comparison with two winding transformers - Open Circuit and Short Circuit tests – Sumpner’s test – Parallel operation with equal and unequal voltage ratios.
UNIT-V (10 Hrs)	Three-Phase Transformers: Polyphase connections- Y/Y, Y/ Δ , Δ /Y, Δ / Δ , open Δ and vector groups, Scott connection – Parallel operation–three winding transformers- off load and on load tap changers.
Textbooks:	
1.	Kothari.D.P and Nagrath.I.J., “Electrical machines”, McGraw Hill Education; 4 th edition,2010.
2.	Bimbhra.P.S, Electrical Machinery, Khanna Publishers, 2011.
Reference Books:	
1.	Mg Say, Theory, “Performance & Design of A.C Machines”, CBS publishers.
2.	Irving L. Kosow, “Electrical Machines & Transformers”, Prentice Hall; 2nd Revised edition 1990.
3.	Hill Stephen, Chapman.j, “Electric Machinery Fundamentals”, McGraw-Hill HigherEducation, 4 edition (2004).
e-Resources	
1.	https://nptel.ac.in/courses/108105112
2.	https://nptel.ac.in/courses/108105155

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE2103	PC	3	--	--	3	30	70	3 Hrs.
ANALOG ELECTRONIC CIRCUITS								
(For EEE)								
Course Objectives: Student will learn								
1.	To acquire the basic knowledge on the h-parameters of a transistor circuit & biasing circuits.							
2.	To understand the concepts of feedback amplifiers.							
3.	To know the operation of oscillators and operational amplifier.							
4.	To understand the applications of operational amplifier.							
5.	To acquire the knowledge on IC 555 timer and their applications.							
Course Outcomes: At the end of the course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Use small signal modeling for transistor circuit analysis. Understand different types of biasing circuits of a transistor.							K2
2.	Illustrate the operation of feedback amplifiers.							K2
3.	Explain the operation of oscillators, operational amplifier and their applications.							K3
4.	Explain 555 timers functionality for the application of multi-vibrators.							K3
5.	Describe the operation of different ADC's and DAC's.							K3
SYLLABUS								
UNIT-I (10 Hrs)	DC biasing of BJTs: Operating Point, Bias Stability, Collector-to-Base Bias, Self-Bias, Stabilization against Variations in V_{BE} and β for the Self-Bias Circuit, Bias Compensation. Small Signals Modelling of BJT: Analysis of a Transistor Amplifier Circuit using h-parameters, Simplified CE Hybrid Model, Frequency Response of CE amplifier.							
UNIT-II (12Hrs)	Feedback Amplifiers: Classification of Amplifiers, the Feedback Concept, General Characteristics of Negative-Feedback Amplifiers, Effect of Negative Feedback upon Output and Input Resistances, Voltage-Series Feedback, Current-Series Feedback, Current-Shunt Feedback, Voltage-Shunt Feedback, Comparison of topologies.							
UNIT-III (10Hrs)	Oscillator Circuits: Barkhausen Criterion of oscillation, Oscillator operation, R-C phase shift oscillator, Wien bridge Oscillator, Crystal Oscillator.							

	Operational Amplifiers: Introduction, Ideal Operational Amplifier, Block Diagram Representation of Typical Op-Amp, OP-Amps Characteristics: Introduction, DC and AC characteristics, 741 op-amp & its features.
UNIT-IV (8Hrs)	OP-AMPS Applications: Introduction, Basic Op-Amp Applications, Instrumentation Amplifier, AC Amplifier, V to I and I to V Converter, Differentiator, integrator. Comparators and Waveform Generators: Introduction, Comparator, Square Wave Generator, Monostable Multivibrator.
UNIT-V (8 Hrs)	Timers: Introduction to 555 timer, functional diagram, Monostable and astable operations. Digital to Analog and Analog to Digital Converters: Introduction, basic DAC techniques, weighted resistor DAC, R-2R ladder DAC, A-D Converters –successive approximation ADC and dual slope ADC.
Textbooks:	
1.	Electronic Devices and Circuit Theory – Robert L.Boylestad and Lowis Nashelsky, Pearson Edition, 2021.
2.	Linear Integrated Circuits – D. Roy Choudhury, New Age International (p) Ltd, 2 nd Edition, 2003.
Reference Books:	
1.	Electronic Devices and Circuits- J. Millman, C.Halkias, Tata Mc-Graw Hill, 2 nd Edition, 2010.
2.	Electronic Devices and Circuits – David Bell, Oxford, 5 th Edition, 2008.
3.	Electronic Devices and Circuits–G.K. Mithal, Khanna Publisher, 23 rd Edition, 2017.
4.	Electronic Principles–Malvino, Albert Paul, and David J. Bates, McGraw-Hill/Higher Education, 2007.
5.	Operational Amplifiers and Linear Integrated Circuits– Gayakwad R.A, Prentice Hall India, 2002.
6.	Operational Amplifiers and Linear Integrated Circuits –Sanjay Sharma, Kataria& Sons, 2 nd Edition, 2010
e-Resources	
1.	https://nptel.ac.in/courses/122106025
2.	https://nptel.ac.in/courses/108102112

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE2104	PC	--	--	3	1.5	30	70	3 Hrs.
ELECTRICAL MACHINES-I LABORATORY								
(For EEE)								
Course Objectives: Students will conduct								
1	Hopkinson’s test, load (brake) test on DC shunt motor and DC series motor.							
2	OC &SC tests and load test on single phase transformer.							
3	Swinburne’s test and Sumpner’s test.							
4	Open Circuit Characteristics (OCC)and Load characteristics of DC generator.							
5	Speed control of DC shunt motors.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1	Analyze the performance of DC Shunt and DC series motors by Conducting Load tests on DC motors and Hopkinson’s test.							K4
2	Analyze the regulation and efficiency of the transformer by Conducting OC and SC tests and Load test on transformer.							K4
3	Calculate the Efficiency of DC shunt machine and Transformer by Conducting Swinburne’s test and Sumpner’s test.							K4
4	Obtain the OCCand load characteristics of DC generator (Shunt and compound) experimentally.							K4
5	Analyze speed characteristics of a DC shunt motor experimentally.							K4
LIST OF EXPERIMENTS								
1	Swinburn’s Test							
2	Speed control of a DC shunt motor.							
3	Load (Brake) test on DC Shunt motor.							
4	Load (Brake) test on DC series motor.							
5	Load test on DC Compound generator.							
6	Open circuit characteristics of a DC shunt generator.							
7	Hopkinson's Test.							
8	Internal and external characteristics of a DC shunt generator.							
9	OC and SC tests on a single-phase transformer.							
10	Load test on a single-phase transformer.							
11	Sumpner's Test.							
12	Scott Connection							

Add on Experiments	
1	Characteristics of BLDC motor.
Reference Books:	
1	Electrical Machine by Dr. P. S. Bhimbra Khanna publications, 2011.
2	Irving L. Kosow, “Electrical Machines & Transformers”, Prentice Hall; 2nd Revised Edition 1990.
3	Clayton, A.E; “Performance and Design of direct machines” CBS; IST edition (2004).



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE2105	PC	--	--	3	1.5	30	70	3 Hrs.
ANALOG ELECTRONIC CIRCUITS LABORATORY								
(For EEE)								
Course Objectives: Students will learn								
1.	To illustrate the concepts of biasing in BJT.							
2.	To analyze single stage amplifier circuits using equivalent circuits.							
3.	To empower students to understand the design and working of BJT amplifiers, oscillators and Operational Amplifiers.							
4.	To get practical experience in design, assembly and evaluation of Linear IC's & its applications. They will use Multisim to test their electronic designs.							
Course Outcomes: At the end of the course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Analyze the performance of BJT amplifier circuits.							K4
2.	Apply the concepts of amplifier analysis to verify their characteristics and measure the important parameters.							K3
3.	Simulation and Design of different amplifiers and oscillator circuits.							K4
LIST OF EXPERIMENTS (Minimum of Ten Experiments has to be performed)								
1.	Frequency Response of RC Coupled Amplifier.							
2.	Frequency Response of Negative Feedback Amplifier.							
3.	Colpitts Oscillator.							
4.	Hartley Oscillator.							
5.	Wein Bridge Oscillator.							
6.	RC Phase Shift Oscillator.							
7.	Basic Applications of Operational Amplifier.							
8.	IC 555 Timer – Astable Multivibrator Circuits.							
9.	Frequency Response of RC Coupled Amplifier using Multisim.							
10.	Frequency Response of Negative Feedback Amplifier using Multisim.							
11.	Basic Applications of Operational Amplifier using Multisim.							
12.	IC 555 Timer – Astable Multivibrator Circuits with Multisim.							
Reference Books:								
1.	Lab manual							

e-Resources:

1.	https://nptel.ac.in/courses/117/103/117103063/
2.	https://www.orcad-.com/resources/orcad-tutorials
3.	https://nptel.ac.in/courses/108/108/108108114/



SRKR
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Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE2106	SEC	--	1	2	2	30	70	3 Hrs.
CIRCUIT SIMULATION LABORATORY								
(For EEE)								
Course Objectives: Students will learn								
1	To solve linear and non-linear equations for electrical circuits.							
2	To solve the differential equations and resonant frequency for electrical circuits.							
3	To obtain the transient response of RL, RC and RLC networks.							
4	To verify the theorems and obtain the parameters of two - port network using simulation.							
5	About the performance characteristics of DC Shunt motor, three phase balanced circuits and solar photovoltaic (PV) system using simulation.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1	Solve the simultaneous linear and non-linear equation for electrical circuits using MATLAB Programming.							K4
2	Solve the differential equations and obtain the frequency response for electrical circuits using MATLAB Programming.							K4
3	Demonstrate the transient response of RL, RC and RLC networks using SIMULINK.							K3
4	Analyze the electrical circuits using theorems and obtain the parameters of two-port network using SIMULINK.							K4
5	Obtain the performance characteristics of DC Shunt motor, three phase balanced circuits and Solar PV system using SIMULINK.							K4
LIST OF PROGRAMMES/ EXPERIMENTS								
1	Solve the simultaneous equations for electrical circuits by Crammer's rule using MATLAB program.							
2	Solve the simultaneous equations for electrical circuits by Gauss Elimination method using MATLAB program.							
3	Obtain iterative solutions for non-linear equations by G-S method & N-R method using MATLAB program.							
4	Solve the differential equations for electrical circuits by Runge - Kutta (RK) method using MATLAB program.							
5	Obtain the Resonant Frequency, Bandwidth and Quality factor for a given series and parallel RLC circuit using MATLAB program.							
6	Obtain the transient response for a given series RL, RC and RLC network using Simulink.							

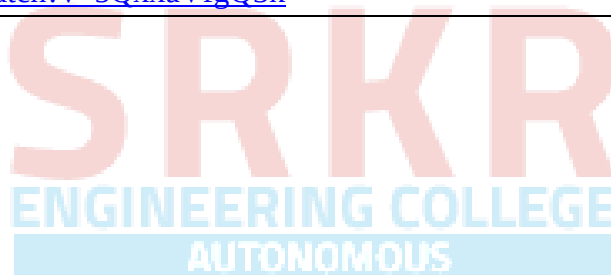
7	Verification of Superposition and Maximum Power Transfer theorem using Simulink.
8	Verification of Thevenin's theorem for AC circuits using Simulink.
9	Determine Z, Y, H and ABCD parameters of a two-port network using Simulink.
10	Obtain the performance characteristics of DC shunt motor using Simulink.
ADD ON EXPERIMENTS	
11	Measurement of Active power and Reactive Power for three phase balanced star connected networks using Simulink
12	Plot I-V & P-V characteristics for Solar PV system using Simulink
Reference Books:	
1	W.H. Hayt & J.K. Kemmerly and Steven M. Durbin, "Engineering circuit analysis", Tata McGraw Hill, 9th edition, New Delhi, 2019.
2	Sudhakar A and Shyam Mohan SP, "Circuits and Networks: Analysis and Synthesis", Tata McGraw Hill, 5 th edition, New Delhi, 2017.
3	John O. Attia, "Electronics and Circuit Analysis Using MATLAB", CRC Press Boca Raton, 2 nd edition, London New York Washington, D.C, 2004.



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC2102	MC	2	--	--	--	30	--	--
ENVIRONMENTAL SCIENCE								
(Common to CIC, CSG, CSE, ECE and EEE)								
Course Objectives: The objective of the course is to impart								
1.	Overall view on natural resources.							
2.	Awareness on ecosystem and its services.							
3.	Various environmental challenges induced due to unplanned anthropogenic activities.							
4.	Consciousness on the social issues, environmental legislation and global treaties							
Course Outcomes: At the end of the course, the students will be able to								
S. No	Outcome							Knowledge Level
1	Describe natural resources and their interaction							K2
2	Illustrate ecosystem types, biodiversity and conservation strategies							K2
3	Summarize contaminants of environment and preventive methods							K2
4	Explain protection of environment by employing constitutional provisions							K2
5	Explain global scenario of surroundings and social conditions							K2
SYLLABUS								
UNIT-I (8 Hrs)	Multidisciplinary Nature of Environmental Studies: Definition, Scope and Importance – Need for Public Awareness. Natural Resources: Renewable and non-renewable resources – Natural resources and associated problems. Forest resources – Use and over – exploitation, deforestation, case studies – Timber extraction – Mining, dams and other effects on forest and tribal people. Water resources – Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems. Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies. Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. Energy resources: Growing energy needs, renewable and non-renewable energy sources use of alternate energy sources							
UNIT-II (8 Hrs)	Ecosystems: Concept of an ecosystem. – Structure and function of an ecosystem – Producers, consumers and decomposers – Energy flow in the ecosystem – Ecological succession – Food chains, food webs and ecological pyramids – Introduction, types, characteristic features, structure and function of the following ecosystem: a. Forest ecosystem. b. Grassland ecosystem c. Desert ecosystem d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)							

	Biodiversity and Its Conservation : Introduction and Definition - genetic, species and ecosystem diversity – Bio-geographical classification of India – Value of biodiversity: consumptive use, Productive use, social, ethical, aesthetic and option values – Biodiversity at global, National and local levels – India as a mega-diversity nation – Hot-spots of biodiversity – Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – Endangered and endemic species of India – Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.
UNIT-III (6 Hrs)	Environmental Pollution: Definition, Cause, effects and control measures of: a. Air Pollution. b. Water pollution c. Soil pollution d. Marine pollution e. Noise pollution f. Thermal pollution g. Nuclear hazards. Solid Waste Management: Causes, effects and control measures of urban and industrial wastes – Role of an individual in prevention of pollution – Pollution case studies – Disaster management: floods, earthquake, cyclone and landslides
UNIT-IV (6 Hrs)	Social Issues and the Environment: From Unsustainable to Sustainable development – Urban problems related to energy watershed management – Resettlement and rehabilitation of people; its problems and concerns – Carbon credits, Mission LiFE. Environmental ethics: Issues and possible solutions. Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust –Wasteland reclamation – Consumerism and waste products. Environment Protection Act. – Air (Prevention and Control of Pollution) Act. – Water (Prevention and control of Pollution) Act – Wildlife Protection Act – Forest Conservation Act. Issues involved in enforcement of environmental legislation – Public awareness.
UNIT-V (6 Hrs)	Human Population And The Environment: Population growth, variation among nations. Population explosion – Family Welfare Programmes. – Environment and human health – Human Rights – Value Education – HIV/AIDS – Women and Child Welfare – Role of information Technology in Environment and human health – Case studies. Field Work: Visit to a local area to document environmental assets River/forest grassland/hill/mountain – Visit to a local polluted site-Urban/Rural/Industrial/Agricultural Study of common plants, insects, and birds – river, hill slopes, etc.
Text Books:	
1.	Erach Bharucha, Text book of Environmental Studies for Undergraduate Courses, Universities Press (India) Private Limited, 2019.
2.	Palaniswamy, Environmental Studies, 2/e, Pearson education, 2014.
3.	S. Azeem Unnisa, Environmental Studies, Academic Publishing Company, 2021.
4.	K. Raghavan Nambiar, “Text book of Environmental Studies for Undergraduate Courses as per UGC model syllabus”, SciTech Publications (India), Pvt. Ltd, 2010.
5.	K. V. S. G. Murali Krishna, The Book of Environmental Studies, Savera Publishing House.
6.	Environmental Studies, R. Rajagopalan, 2 nd Edition, 2011, Oxford University Press.

Reference Books:	
1.	Deeksha Dave and S.S. Katewa, Textbook of Environmental Studies, 2/e, Cengage Learning.
2.	M. Anji Reddy, “Textbook of Environmental Sciences and Technology”, BS Publication, 2014.
3.	J.P. Sharma, Comprehensive Environmental studies, Laxmi publications, 2006.
4.	J. Glynn Henry and Gary W. Heinke, Environmental Sciences and Engineering, Prentice Hall of India Private limited, 1988.
5.	G.R. Chatwal, A Text Book of Environmental Studies, Himalaya Publishing House, 2018
6.	Gilbert M. Masters and Wendell P. Ela, Introduction to Environmental Engineering and Science, 1/e, Prentice Hall of India Private limited, 1991.
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc23_hs155/preview
2.	https://www.edx.org/learn/environmental-science/rice-university-ap-r-environmental-science-part-3-pollution-and-resources?index=product&objectID=course-3a6da9f2-lec07.pdf (iasri.res.in)
3.	http://ecoursesonline.iasri.res.in/Courses/Environmental%20Science-I/Data%20Files/pdf/lec07.pdf
4.	https://www.youtube.com/watch?v=5QxxaVfgQ3k





SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		II / IV - B.Tech. II - Semester							
ELECTRICAL & ELECTRONICS ENGINEERING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23HS2201	Managerial Economics and Financial Analysis	HS	2	0	0	2	30	70	100
B23EE2201	Electrical Measurements & Instrumentation	ES	3	0	0	3	30	70	100
B23EE2202	Electromagnetic Field Theory	PC	2	1	0	3	30	70	100
B23EE2203	Induction & Synchronous Machines	PC	2	1	0	3	30	70	100
B23EE2204	Digital Electronics and Logic Design	PC	3	0	0	3	30	70	100
B23EE2205	Electrical Machines – II Laboratory	PC	0	0	3	1.5	30	70	100
B23EE2206	Digital Design Laboratory	PC	0	0	3	1.5	30	70	100
B23EE2207	Smart Systems Laboratory	SEC	0	1	2	2	30	70	100
B23EE2208	Design Thinking & Innovation	ES	1	0	2	2	30	70	100
B23MC2201	English Proficiency	MC	2	0	0	-	30	-	30
TOTAL			15	3	10	21	300	630	930

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS2201	HS	2	--	--	2	30	70	3Hrs.
MANAGERIAL ECONOMICS AND FINANCIAL ANALYSIS								
(Common to AIDS, CSE, CIC, CSG, CSIT, CE, ECE, EEE, ME)								
Course Objectives: Students are expected to								
1.	Understand the concept and nature of Managerial Economics, its relationship with other disciplines, the Concept of Demand and Demand forecasting							
2.	Familiarize about the Concepts of Cost and Break-Even Analysis							
3.	Learn about accounting cycle and preparation of Financial Statements							
4.	Understand the nature of markets and the Pricing Methods							
5.	Know the concept of Capital, Sources of Raising Finance and Depreciation							
Course Outcomes: At the end of the course the student will be able to								
S.No	Outcome							Knowledge Level
1.	Interpreting the importance of Managerial Economics, demand analysis and methods of demand forecasting							K2
2.	Describe about the usefulness of Cost Analysis and Break Even Analysis							K2
3.	Apply the principles of accounting to convert the transactions and events into Journal, Ledger and Trail balance							K3
4.	Compute the results of Business by preparing Final Accounts							K3
5.	Illustrate the nature of markets and pricing theories							K2
6.	Explain the Types of capital, their sources and importance & estimation of Depreciation							K2
SYLLABUS								
UNIT-I (12 Hrs)	Introduction to Managerial Economics and demand Analysis: Managerial Economics: Definition of Economics & Classification of Economics (Micro & Macro), Meaning, Nature, & Scope of Managerial Economics. Demand Analysis: Concept of Demand, Determinants of Demand, Demand schedule, Demand curve, Law of Demand and its exceptions. Elasticity of Demand, Types of Elasticity of Demand. Importance of Demand Forecasting and its Methods.							
UNIT-II (8 Hrs)	Cost Analysis: Importance of cost analysis, Types of Cost- Actual cost Vs Opportunity cost, Fixed cost Vs Variable cost, Explicit Vs Implicit cost, Historical cost Vs Replacement cost, Incremental cost Vs Sunk cost; Elements of costs – Material, Labour, Expenses; Methods of costing - Job costing, contract costing, Process costing, Batch costing, Unit costing, Service costing, Multiple costing. Break-even analysis: Determination of Break-							

	even Point Applications, Assumptions and Limitations of Break-even analysis (Theory only).
UNIT-III (12 Hrs)	Introduction to Financial Accounting: Importance of Accounting - Double Entry System of Accounting - Types of Accounts - Journal, Ledger, Trail Balance, Trading Account, Profit and Loss Account and Balance Sheet (Final Accounts with Simple adjustments).
UNIT-IV (8 Hrs)	Introduction to Markets & Pricing Policies Market Structures: Salient Features of Perfect Competition, Monopoly, Monopolistic competition, Oligopoly and Duopoly. Pricing: Importance of pricing and its meaning; Methods of Pricing: Cost Based -Full cost, Mark-up, Marginal & Break-even; Demand Based -Penetrating, Skimming; Competition Based -Going rate, Sealed Bid, Discount; Internet Pricing -Flat-rate, Usage sensitive.
UNIT-V (8 Hrs)	Capital & Depreciation: Types of Capital-Fixed capital & Working Capital, Components of Working Capital, Factors influencing Working capital. Methods of Raising Finance - Short term, Medium term and Long term. Depreciation – Meaning, Importance and causes of depreciation; Methods of Depreciation-Straight line and Diminishing Balancing methods (Theory only).
Text Books:	
1.	AR Aryasri, Managerial Economics and Financial Analysis, TMH Pvt. Ltd, New Delhi
2.	Dr. N. Appa Rao, Dr. P. Vijaya kumar: Managerial Economics and Financial Analysis, Cengage Publications, NewDelhi
Reference Books:	
1.	Dr.B.Kuberudu & T.V.Ramana :Managerial Economics and Financial anaysis, Himalaya Publishing House
2.	Varshney R.L, K.L Maheswari, Managerial Economics, S. Chand & Company Ltd,
3.	Shashi K.Gupta & R.K.Sharma Management Accounting, Kalyani Publishers
4.	Maheswari S.N, An Introduction to Accountancy, Vikas Publishing House Pvt Ltd

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23EE2201	ES	3	--	--	3	30	70	3 hrs.
ELECTRICAL MEASUREMENTS AND INSTRUMENTATION								
(For EEE)								
Course Objectives: Students will learn								
1.	The operation of moving coil, moving iron and dynamometer type instruments for measuring electrical quantities.							
2.	About the compensation techniques for accurate measurement and extension of range of measurement.							
3.	The usage of different bridges for the measurement of Capacitance, Inductance, Frequency and Instrument Transformers.							
4.	The operation and usage of Digital meters, Cathode Ray Oscilloscope (CRO) and Digital Storage Oscilloscope (DSO).							
5.	The operation of different transducers for measuring non-electrical quantities.							
Course Outcomes: At the end of the course, students will be able to								
S. No	Outcome							Knowledge Level
1.	Illustrate the operation of moving coil, moving iron and dynamometer type instruments for measuring electrical quantities.							K3
2.	Apply the compensation techniques for accurate measurement and extension of range of measurement.							K3
3.	Apply the usage of different bridges for the measurement of Capacitance, Inductance & Frequency and illustrate the Instrument Transformers.							K3
4.	Examine the operation and usage of Digital meters, Cathode Ray Oscilloscope and Digital Storage Oscilloscope.							K3
5.	Illustrate the operation of different transducers for measuring non-electrical quantities.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Measuring Instruments: Classification, Error Analysis, Deflecting, Controlling and Damping torques, Moving Coil, Moving iron type instruments, Expression for the torque, Errors, and compensations. Extension of range using shunts and multipliers.							
UNIT-II (10 Hrs)	Measurement of Power and Energy: Dynamometer type wattmeter - Torque expression, Errors. Power in Poly-phase systems -Two wattmeter method, Effect of Power factor on wattmeter reading. Measurement of Reactive power. Power factor meter, Energy meter - Torque expression. Introduction to							

	Digital Energy meter.
UNIT-III (10 Hrs)	AC Bridges & Instrument Transformers Methods of Measuring Inductance, Capacitance and Frequency-Maxwell's Bridge, Hay's Bridge, Schering Bridge, Wein's Bridge. Instrument Transformers - Current Transformer, Phasor Diagram, Errors; Potential Transformer - Phasor Diagram, Errors.
UNIT-IV (10 Hrs)	Digital meters and Cathode Ray Oscilloscope (CRO) Digital Voltmeter - Ramp type DVM- Dual-slope integrating type - Successive-approximation type. Digital Multimeter. CRO - Block Diagram - Cathode Ray Tube (CRT), Measurement of Voltage frequency and Time period, Lissajous Patterns. Digital Storage Oscilloscope (DSO).
UNIT-V (10 Hrs)	Measurement of Non-Electrical Quantities Transducers – Classification, Characteristics of Transducers. Measurement of Pressure - Strain Gauge, Piezo-Electric Transducers; Measurement of Displacement – Linear Variable differential transformer (LVDT); Measurement of Temperature -Thermocouple. Electromagnetic Flowmeter, Light Dependent Resistor (LDR).
Text Books:	
1.	Sawhney, . A.K., “A Course in Electrical & Electronic Measurement and Instrumentation,” Dhanpat Rai & Company Private Limited, New Delhi, 2023, ISBN: 9788121997751
2.	R.S. Sedha “Electronic Measurements and Instrumentation,” S. Chand & Company Pvt. Ltd, New Delhi, 1 st Edition, 2013, ISBN: 81-219-9775-5
Reference Books:	
1.	Rangan, C.S., Sharma, G.R., Mani, V.S., “Instrumentation Devices and Systems,” Tata McGraw- Hill Publishing Company, New Delhi, 2 nd Edition, 2002.
2.	H S Kalsi “Electronic Instrumentation”, Tata McGraw-Hill Publishing Company, New Delhi, 4 th Edition, 2019.ISBN-10:978-93-5316-251-3
e-Resources:	
1.	https://onlinecourses.nptel.ac.in/noc19_ee44 (Electrical Measurement and Electronic Instruments)
2.	https://nptel.ac.in/courses/108102191 (Transducers For Instrumentation)

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE2202	PC	2	1	--	3	30	70	3 Hrs.
ELECTRO MAGNETIC FIELD THEORY								
(For EEE)								
Course Objectives: Students will learn								
1.	The basic laws and theorems that calculate electric field and forces.							
2.	The concepts about the effects of electric fields on conductors, insulators and their boundary conditions.							
3.	The basic laws and theorems that calculate magnetic fields.							
4.	About the properties of magnetic materials, their boundary conditions, magnetic potential & energy.							
5.	Maxwell's equations and poynting theorem.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Compute the electrostatic forces and electric field intensity for given charge configuration by applying Vector Calculus and Coulomb's Law.							K3
2.	Calculate the Electric Potential, Energy and discriminate the conductors and dielectrics by its properties.							K3
3.	Calculate the Magnetic field intensity by using Biot-Savart's, Ampere's law.							K3
4.	Calculate the Magnetic potential, energy & Illustrate the properties of magnetic materials.							K3
5.	Illustrate Maxwell's equations (using Faraday's, Lenz's laws), poynting vector and poynting theorem.							K3
SYLLABUS								
UNIT-I (10Hrs)	Vector Calculus: Review of Scalars & vectors, unit vector, vector operations, Coordinate Systems: Review of Rectangular, cylindrical and spherical coordinate systems, Electrostatics: Coulomb's law and superposition principle, different types of charge configurations, electric flux, electric field intensity and electric flux density, electric field intensity and electric flux density due to different charge configurations.							
UNIT-II (10 Hrs)	Electric potential: Gauss's law in integral form and point form in terms of D, applications of Gauss' law, Divergence theorem, Electric potential, calculation of electric potential, electrostatic energy, Electrostatic boundary conditions, basic properties of conductors in electrostatic fields, capacitance, Poissons and Laplace's equations, solutions of Laplace's equations for							

	single variables, electric dipoles, polarization of dielectrics.
UNIT-III (10 Hrs)	Magneto-statics: Biot-savart's law, determination of magnetic field intensity and magnetic flux density due to various steady current configurations, curl of H, Ampere's circuits law in integral and differential form, applications of Ampere's law, Stokes theorem.
UNIT-IV (10 Hrs)	Magnetic Potential: Calculation of scalar and vector magnetic potential, Magneto-static boundary conditions. Lorentz force equation, Force between two straight long and parallel current carrying conductors. The magnetic dipole, magnetization, bound current, inductance and energy in magnetic fields.
UNIT-V (10 Hrs)	Time varying fields and Maxwell's equations: Faraday's laws, Lenz's law, Maxwell's equations, statically and dynamically induced EMF, modification of ampere's circuital law for time varying fields – displacement current and current density, the poynting vector, poynting theorem.
Textbooks:	
1.	Engineering electromagnetics by William H. Hayt, John A. Buck McGraw-Hill Publishing Co., 8 th edition.
2.	Principles of Electromagnetics by Mathew N.O. Sadiku, Oxford; 6 th edition
Reference Books:	
1.	Introduction to electro dynamics by D.J. Griffiths, PHI Learning; 3 rd Edition (2012).
e-Resources	
1.	https://nptel.ac.in/courses/108/106/108106073/#
2	https://nptel.ac.in/courses/108104087

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE2203	PC	2	1	--	3	30	70	3 Hrs.
INDUCTION AND SYNCHRONOUS MACHINES								
(For EEE)								
Course Objectives: Students will learn about								
1.	The construction and operation of three-phase induction motors.							
2.	The torque production and performance of three-phase induction Motors.							
3.	The construction and working of single-phase motors.							
4.	The working and performance of synchronous generator.							
5.	The working and performance of synchronous motor.							
Course Outcomes: At the end of the course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Explain the construction, operation, equivalent circuit and examine the performance of three-phase induction motor.							K3
2.	Analyze the performance of three-phase induction motor.							K4
3.	Illustrate the construction, working and starting methods of single-phase motors.							K3
4.	Analyze the performance of Synchronous generators.							K4
5.	Analyze the performance of Synchronous motors.							K4
SYLLABUS								
UNIT-I (10Hrs)	Three-phase induction motors: Construction of Squirrel cage and Slipring induction motors– production of rotating magnetic field – principle of operation – slip and slip frequency – rotor emf, equivalent circuit- phasor diagram- rotor current and power factor at standstill and during running conditions–Power flow diagram, rotor power input, rotor copper loss and mechanical power developed and their inter-relationship.							
UNIT-II (10 Hrs)	Performance of Three-Phase induction motors: Torque equation – expressions for maximum torque and starting torque – torque-slip characteristics –No load test, Blocked rotor test and Brake test – circle diagram - methods of starting–starting current and torque calculations -speed control of induction motor with V/f control, rotor resistance control and rotor emf injection technique –crawling and cogging – induction generator operation.							

UNIT-III (10 Hrs)	Single Phase Motors: Single phase induction motors – constructional features – double field revolving theory, equivalent circuit- starting methods: capacitor start capacitor run, capacitor start induction run, split phase & shaded pole, AC series motor and applications.
UNIT-IV (10 Hrs)	Synchronous Generators (Alternators): Constructional features of salient pole and non-salient pole synchronous generators- armature windings –concentrated and distributed windings, Short pitched coils – distribution factor & pitch factor – E.M.F equation –armature reaction – voltage regulation by synchronous impedance method, MMF method and Potier triangle method –two reaction theory, analysis of salient pole machines - Slip test, methods of synchronization- Parallel operation of alternators.
UNIT-V (10 Hrs)	Synchronous Motors: Synchronous motor principle and theory of operation – Operating characteristics of synchronous motor- variable excitation and constant load, V & inverted V curves, variable load and constant excitation – synchronous condenser –expression for power developed – hunting and its suppression –starting methods of synchronous motors.
Textbooks:	
1.	Electrical Machinery, Dr. P.S. Bhimbra, Khanna Publishing, 2021, 1 st Edition.
2.	Electrical machines, D.P. Kothari and I.J. Nagrath, McGraw Hill Education, 2017, 5 th Edition.
Reference Books:	
1.	Performance and analysis of AC machines by M.G. Say, CBS, 2002.
2.	Theory & Performance of Electrical Machines by J.B.Gupta, S.K.Kataria & Sons, 2007.
3.	Electric Machinery, A.E.Fitzgerald, Charles Kingsley, Stephen D.Umans, McGraw-Hill, 2020, 7 th edition.
e-Resources	
1.	https://nptel.ac.in/courses/108/105/108105131
2.	https://nptel.ac.in/courses/108106072

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE2204	PC	3	--	--	3	30	70	3 Hrs.
DIGITAL ELECTRONICS AND LOGIC DESIGN								
(For EEE)								
Course Objectives: Students will learn about								
1.	The implementation of Combinational logic circuits.							
2.	The Flipflops and design of Sequential logic circuits							
3.	The characteristics of digital integrated circuits and memory implementation using programmable logic technologies such as RAM, ROM, PLA, and PAL.							
4.	The Verilog HDL to implement combinational circuits.							
5.	The Verilog HDL to implement sequential circuits.							
Course Outcomes: At the end of the course, the students will be able to								
S. No	Outcome							Knowledge Level
1.	Implement combinational logic circuits using K – maps.							K3
2.	Implement Flipflops using gates and design shift registers & counters.							K3
3.	Examine the digital integrated circuit characteristics and implement PLA, PAL circuits.							K3
4.	Model the combinational logic circuits using Verilog HDL.							K3
5.	Model the sequential logic circuits using Verilog HDL.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Combinational Logic Circuits: Function minimizations, Three & Four Variable K-maps, POS Simplifications, don't care conditions, NAND and NOR Implementation, AND-OR-INVERT implementation. Analysis and Design procedure of combinational circuits-Half adder, Full adder, Binary adder, Binary subtractor, DeCourse Coder, EnCourse Coder, Multiplexer, Demultiplexers.							
UNIT-II (10 Hrs)	Sequential Logic Circuits: Introduction to Sequential circuits, Flip flops – RS, D, JK and T logic diagrams, operations, characteristic and excitation tables, Master-Slave JK flip-flop, Sequential circuits design procedure- Shift register, Universal shift register, Binary Ripple counter, Synchronous counters, Up - Down synchronous counters.							

UNIT-III (10Hrs)	Digital Integrated Circuits: Introduction, Propagation delay, Power dissipation, Fan-out and Fan-in, Noise margin, Bipolar Transistor Characteristics, RTL, DTL, TTL, ECL and CMOS. Memory And Programmable Logic: Introduction, RAM, ROM, Types of ROM, Programmable Logic Array (PLA), Programmable Array Logic (PAL).
UNIT-IV (10 Hrs)	Combinational circuit design using Verilog HDL Introduction to Hardware description language (HDL), Design of digital integrated circuits, Module declaration, Propagation delays, Stimulus, Boolean equations, User defined Primitives, HDL Models of Combinational circuits- Gate level modelling, Dataflow modelling, Behavioral modelling.
UNIT-V (10 Hrs)	Sequential circuits design using Verilog HDL HDL Models of Sequential Circuits- Blocking and non-blocking statements, HDL program for the given state diagram, HDL Model of universal shift registers, HDL Model of 4-bit ripple and synchronous counters.
Textbooks:	
1.	M. Morris Mano & Michael D. Ciletti, "Digital Design with an introduction to the Verilog HDL, VHDL and system Verilog" Pearson India educations services Pvt. Limited, 6th Edition, 2018.
2.	R. Ananda Natarajan, "Digital Design", Prentice-Hall of India Pvt. Limited, New Delhi, 2015.
3.	Stephen Brown & Zvonko Vranesic, "Fundamentals of Digital logic design with Verilog HDL", Tata McGraw Hill, 3rd edition.
Reference Books:	
1.	Frederick J. Hill and Gerald R. Peterson, Introduction to Switching Theory and Logic Design, John Wiley & sons, Inc. New York, 3 rd edition, 1981.
2.	Michael D. Ciletti, "Advanced digital design with the Verilog HDL", Eastern Economy Edition, PHI, 2 nd edition, 2010.
3.	Bhaskar, "VHDL Primer", 3 rd Edition, PHI Publications, 2015.
4.	Charles H Ruwth, "Fundamentals of Logic design" 7 th edition, CI-Engineering publishers, 2013.
e-Resources	
1.	https://nptel.ac.in/courses/117/106/117106086/
2.	https://onlinecourses.nptel.ac.in/noc24ee94/course

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE2205	PC	--	--	3	1.5	30	70	3 Hrs.
ELECTRICAL MACHINES-II LABORATORY								
(For EEE)								
Course Objectives: Students will learn								
1	To find voltage regulation of an alternator by using EMF, MMF and ZPF methods.							
2	To conduct tests online excited induction generator and synchronous motor.							
3	To conduct No Load and Blocked Rotor tests on a 3-Φ and 1-Φ induction motor.							
4	To conduct test on synchronous machine to find X _d and X _q and sequence reactances.							
5	To conduct Load test on 3-Φ induction motor.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1	Determine the voltage regulation of an alternator by using EMF, MMF and ZPF methods by conducting suitable tests.							K4
2	Analyze the performance characteristics of Line excited induction generator and Synchronous motor experimentally.							K4
3	Analyze the performance characteristics of a 3-Φ and 1-Φ Induction motors by conducting No – load and Blocked rotor tests.							K4
4	Determine X _d & X _q and sequence reactances of a Synchronous machine experimentally.							K4
5	Obtain the performance characteristics of three phase slip-ring induction motor experimentally.							K4
LIST OF EXPERIMENTS								
1	Circle diagram of a three-phase Squirrel cage Induction Motor							
2	Voltage regulation of an Alternator by EMF and MMF methods.							
3	Voltage regulation of an Alternator by ZPF method.							
4	Characteristics of line excited Induction Generator.							
5	Characteristics of Induction start Synchronous Motor.							
6	Load test on three phase Slip ring Induction Motor.							
7	V and inverted V curves of a Synchronous Motor.							
8	Measurement of X _d and X _q of a Synchronous Machine.							
9	Equivalent circuit of a single-phase Induction Motor.							
10	Measurement of sequence reactances of a Synchronous Machine.							
11	Speed control of three phase slip-ring induction motor by pole changing method.							
Add on Experiments								

1	Characteristics of Schrage motor.
Reference Books:	
1	Electrical Machinery, Dr. P.S. Bhimbra, Khanna Publishing, 2021, 1 st Edition.
2	Electrical machines, D.P. Kothari and I.J. Nagrath, McGraw Hill Education, 2017, 5 th Edition.



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE2206	PC	--	--	3	1.5	30	70	3 Hrs.
DIGITAL DESIGN LABORATORY								
(For EEE)								
Course Objectives: Students will								
1	Design, implement and evaluate combinational logic circuits, binary adders and Multiplexers using a Digital Trainer Kit.							
2	Design, implement and verify the functionality of synchronous binary counters and shift registers using a Digital Trainer Kit.							
3	Simulate and verify functionality of CMOS inverters, Universal gate sand Boolean expressions using Verilog HDL.							
4	Learn Verilog Behavioral description to implement and verify Course Code converters and enCourse Coders/deCourse Coders.							
5	Simulate and verify the functionality of a D-latch using Verilog HDL.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1	Design the combinational circuits and examine the functionality of Adders and Multiplexers using Digital Trainer Kit.							K4
2	Design the synchronous binary counters and examine the performance of shift registers using the Digital Trainer Kit.							K4
3	Develop Verilog HDL programs to simplify and realize the given Boolean expression and verify the functionality of CMOS inverters, Universal gates.							K4
4	Develop Verilog HDL programs to design and verify the Course Code converters and the functionality of enCourse Coders and deCourse Coders.							K4
5	Develop Verilog HDL program to implement the functionality of a D-latch.							K4
LIST OF EXPERIMENTS								
1	Design of a combinational logic circuit using Digital Trainer Kit.							
2	Implementation of Binary Adders using Digital Trainer Kit							
3	Verify the logic functions of Digital Multiplexer using Digital Trainer Kit.							
4	Design and Verify Synchronous Binary Counter using Digital Trainer Kit.							
5	Verify the functioning of Shift Register using Digital Trainer Kit							
6	Verification of universal gates using Verilog HDL.							
7	To simplify the given Boolean expressions and realize using Verilog program							
8	To realize the following Course Code converters using Verilog Behavioral description a) Gray to binary and vice versa b) Binary to excess3 and vice – versa.							
9	To realize using Verilog Behavioral description 8:1 Multiplexer.							

10	Verification of CMOS inverter using Verilog HDL.
11	Implementation of EnCourse Coder & DeCourse Coder using Verilog HDL
12	Implementation of D-Latch using Verilog HDL
Reference Books:	
1	M. Morris Mano & Michael D. Ciletti, “Digital Design with an introduction to the Verilog HDL, VHDL and system Verilog” Pearson India educations services Pvt. Limited, 6 th Edition, 2018.
2	R. Ananda Natarajan, “Digital Design”, Prentice-Hall of India Pvt. Limited, New Delhi, 2015.
3	Stephen Brown & Zvonko Vranesic, “Fundamentals of Digital logic design with Verilog HDL”, McGraw Hill Higher Education, 3 rd edition, 2013.



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE2207	SEC	--	1	2	2	30	70	3 Hrs.
SMART SYSTEMS LABORATORY								
(For EEE)								
Course Objectives: Students will learn								
1	To study the various sensors used in IoT Networks, Raspberry Pi, Arduino and their peripherals.							
2	To study the wireless communication devices and their communication protocols used in IoT networks.							
3	To study development of IoT application from a global perspective and building automation systems by considering real world constraints.							
4	To understand the implementation of IoT by studying case studies like Smart Home, Smart city, etc.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1	Identity different components of IoT Networks							K2
2	Interface various sensors to Processor boards							K4
3	Apply various IoT network protocols to communicate between sensors and machines wirelessly							K3
4	Connect the devices using web and internet in the IoT environment							K4
5	Develop automation of a system using IoT devices							K4
LIST OF EXPERIMENTS								
1	Familiarization with Arduino/Raspberry Pi and perform necessary software installation. a. To interface LED/Buzzer with Arduino/Raspberry Pi and write a program to turn ON LED for 1 sec after every 2 seconds b. To interface Push button/Digital sensor (IR/LDR) with Arduino/Raspberry Pi and write a program to turn ON LED when push button is pressed or at sensor detection c. To interface Bluetooth with Arduino/Raspberry Pi:							

	i)write a program to send sensor data to a smartphone using Bluetooth. ii)write a program to turn LED ON/OFF when '1'/'0' is received from a smartphone using Bluetooth.
2	a. Write a program to access pwm pin of Arduino and control the LED from variable analog input voltage b. Write a program to map duty cycle values from [0 to 1023] to [0 to 100] at pwm pin of Arduino and show the signal in oscilloscope
3	a. Interfacing of 7 Segment Display with Arduino and write a program to display the hexadecimal number scrolling in 7 segment display b. Interfacing of LCD Display with Arduino and write a program to display and scroll the information in two lines
4	Interface a temperature sensor with Arduino and write a program to print temperature and humidity readings
5	Develop a temperature-based DC motor speed control system using an Arduino board, a DHT11 temperature sensor and display the temperature on an LCD. [The system should be capable of reading the ambient temperature and adjusting the speed of the DC motor accordingly. This experiment mimics the control mechanisms used in home HVAC systems and automotive cooling systems.]
6	Implement a system that controls the speed, direction of rotation, and on/off state of a DC motor using three push buttons with an Arduino microcontroller and an L298N motor driver module.
7	Implement effective control over the speed of a DC motor using an L298N motor driver module and Pulse Width Modulation (PWM) via an Arduino microcontroller and adjust the speed of the motor using a potentiometer.
8	a. Implement the control of a stepper motor's rotation and direction using an Arduino board. b. Implement a system that controls the rotation of a stepper motor based on temperature readings from a sensor. The system should display both the current temperature and the rotation degrees of the motor on an LCD display.
9	Implement a sun tracking system using an Arduino board and stepper motor, enhancing the efficiency of solar energy collection.
10	a. Wirelessly control a servo motor using a Bluetooth module and Arduino board and mobile application. b. Implement a system that controls the position of a servo motor based on the distance measured by an ultrasonic sensor and the measured distance will also be displayed on

	an LCD screen. c. Implement an automatic car wiper system using Arduino and a servo motor, which activates based on moisture sensing.
11	Calibrate current and voltage using appropriate sensors, calculate the electrical power and display the measured values of current, voltage and power on an LCD
12	Implement a home automation system using Arduino to achieve the following functionalities: a. Automatic room lights control using a PIR sensor. b. Automatic temperature-controlled fan using a DHT11 sensor. c. Manual control of lights via a smartphone using an HC-05 Bluetooth module
13	Control of Load Using Relay and Raspberry Pi with Server/Central Command System Design and implement a system to control an electrical load (like a light or a fan) using a relay module interfaced with a Raspberry Pi. The control commands will be issued from a remote server or central command system.
Reference Books and Links:	
1	Vijay Madiseti and ArshdeepBahga, "Internet of Things (A Hands-on-Approach)", 1st Edition,VPT, 2014.
2	Simon Monk, "Programming the Raspberry Pi: Getting Started with Python", January 2012,McGraw Hill Professional Massimo Banzi, "Getting Started with Arduino", 1 st Edition, February 2009, O'Reilly Media, Inc.
3	Olivier Hersent, David Boswarthick, Omar Elloumi, "The Internet of Things – Key Applications and Protocols", Wiley, 2012.
4	Embedded Sensing, Actuation and Interfacing Systems, IIT Kharagpur Prof. Banibrata Mukherjee https://nptel.ac.in/courses/108105376
5	Foundations of Cyber Physical Systems, IIT Kharagpur Prof. Soumyajit Dey https://nptel.ac.in/courses/106105241
6	Introduction to internet of things, IIT Kharagpur Prof. Sudip Misra https://nptel.ac.in/courses/106105166
7	Introduction to Industry 4.0 and Industrial Internet of Things, IIT Kharagpur Prof. Sudip Misra https://nptel.ac.in/courses/106105195

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE2208	ES	1	--	2	2	30	70	3 Hrs.
DESIGN THINKING & INNOVATION								
(Common to all Programmes of Engineering)								
Course Objectives:								
1.	Bring awareness on innovative design and new product development.							
2.	Explain the basics of design thinking.							
3.	Familiarize the role of reverse engineering in product development.							
4.	Train how to identify the needs of society and convert into demand.							
5.	Introduce product planning and product development process.							
Course Outcomes: After completion of this course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Define the concepts related to design thinking.							K1
2.	Explain the fundamentals of Design Thinking and innovation.							K2
3.	Apply the design thinking techniques for solving problems in various sectors.							K3
4.	Analyse to work in a multidisciplinary environment.							K4
5.	Evaluate the value of creativity.							K5
SYLLABUS								
UNIT-I (10Hrs)	Introduction to elements and principles of Design, basics of design-dot, line, shape, form as fundamental design components. Principles of design. Introduction to design thinking, history of Design Thinking, New materials in Industry.							
UNIT-II (10 Hrs)	Design thinking process (empathize, analyze, idea & prototype), implementing the process in driving inventions, design thinking in social innovations. Tools of design thinking - person, costumer, journey map, brainstorming, product development. Activity: Every student presents their idea in three minutes, Every student can present design process in the form of flow diagram or flow chart etc. Every student should explain about product development.							
UNIT-III (10 Hrs)	Art of innovation, Difference between innovation and creativity, role of creativity and innovation in organizations. Creativity to Innovation. Teams for innovation, Measuring the impact and value of creativity. Activity: Debate on innovation and creativity, Flow and planning from idea to innovation, Debate on value-based innovation.							
UNIT-IV (10 Hrs)	Problem formation, introduction to product design, Product strategies, Product value, Product planning, product specifications. Innovation towards product design Case studies.							

	Activity: Importance of modeling, how to set specifications, Explaining their own product design.
UNIT-V (10 Hrs)	Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business – Business challenges: Growth, Predictability, Change, Maintaining Relevance, Extreme competition, Standardization. Design thinking to meet corporate needs. Design thinking for Startups. Defining and testing Business Models and Business Cases. Developing & testing prototypes. Activity: How to market our own product, about maintenance, Reliability and plan for startup.
Textbooks:	
1.	Tim Brown, Change by design, 1/e, Harper Bollins, 2009.
2.	Idris Mootee, Design Thinking for Strategic Innovation, 1/e, Adams Media, 2014.
Reference Books:	
1.	David Lee, Design Thinking in the Classroom, Ulysses press, 2018.
2.	Shrrutin N Shetty, Design the Future, 1/e, Norton Press, 2018.
3.	William lidwell, Kritinaholden, &Jill butter, Universal principles of design, 2/e, Rockport Publishers, 2010.
4.	Chesbrough.H, The era of open innovation, 2003.
e-Resources:	
1.	https://nptel.ac.in/courses/110/106/110106124/
2.	https://nptel.ac.in/courses/109/104/109104109/
3.	https://swayam.gov.in/nd1_noc19_mg60/preview
4.	https://onlinecourses.nptel.ac.in/noc22_de16/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC2201	MC	2	--	--	--	30	--	--
ENGLISH PROFICIENCY								
(Common to CSE, CE, ECE, EEE, CIC, CSD).								
Course Outcomes: After completion of this course, students will be able to								
Sl. No	Outcome							Knowledge Level
1.	Interpret IELTS & TOEFL listening comprehension texts.							K2
2.	Demonstrate essential speaking skills in academic, professional, and real-life contexts.							K2
3.	Interpret the written discourse by applying effective reading strategies.							K2
4.	Construct coherent and cohesive paragraphs, e-mails, letters, and essays.							K3
SYLLABUS								
UNIT-I	Listening Skills Listening as a receptive skill Listening skills for IELTS, TOEFL, Duolingo, PTE, and other competitive examinations.							
UNIT-II	Speaking Skills JAM/ Extempore Debate / Group Discussion Presentation Skills							
UNIT-III	Reading Skills Types of Reading (Intensive and Extensive Reading, Skimming, Scanning) Reading/Summarizing News Paper Articles							
UNIT-IV	Writing Skills Essay Writing (Argumentative, Analytical and Descriptive essays for GRE, TOEFL & IELTS) E-Mail / Letter Writing Resume Writing							
UNIT-V	Integrated Language Skills Listening Skills for Speaking and Writing Reading Skills for Writing and Speaking							
Text Books:								
1.	Interchange (5 th edition) by Jack C.Richards, CUP, 2017.							

Reference Books:	
1.	Fundamentals of Technical Communication (1 st edition) by Meenakshi Raman, Sangeeta Sharma of OUP, 2014.
2.	The Oxford Guide to Writing and Speaking (3 rd edition) by John Seely OUP, 2013.
3.	Effective Technical Communication (2 nd edition) by M. Ashraf Rizwi. TataMcGrawhill, 2017.
e-Resources:	
1	BBC Learning English - Learn English with BBC Learning English - Homepage
2	Grammar Learn English (britishcouncil.org)
3	Duolingo English Test
4	IELTS Test Preparation Materials - Videos, Practice tests, Articles and More (idp.com)

